

Affine-segment preservation on a quarter ball

`MathlibAnnex.IsometryEquiv.map_lineMap_of_mem_ball`

Upgrades local midpoint preservation to preservation of every affine segment parameter inside the quarter-ball chart.

Statement

Under the same ambient-ball hypotheses as the local midpoint theorem, if x and y lie in $\text{ball}(c, R/4)$, then for every a in $[0, 1]$, f sends the affine point $\text{lineMap}(x, y, a)$ to $\text{lineMap}(f(x), f(y), a)$.

Assumptions

Real normed vector spaces; radius- R balls about c and $f(c)$ lie in the source and target; $R > 0$; x, y lie in the radius- $R/4$ source ball; a lies in $[0, 1]$.

Conclusion

f preserves the affine combination $(1-a)x + ay$, with the exact subtype membership witnesses supplied in Lean.

Proof idea

Restrict the domain to the convex quarter ball and view f there as a continuous map into the ambient target space. The preceding local midpoint theorem supplies midpoint preservation on that convex set. A general midpoint-to-line-map lemma then gives preservation of every segment parameter.

Proof steps

1. Define the inclusion of the quarter ball into s and the ambient-valued restriction g of f .
2. Use continuity of the inclusion and of the isometry to prove continuity of g .
3. Apply the quarter-ball midpoint theorem to obtain midpoint preservation for every pair of points in the restricted ball.
4. Invoke the continuous convex-set lemma that upgrades midpoint preservation to affine-line-map preservation on $[0, 1]$.
5. Unfold the local definitions to obtain the stated equality for f .

Main citations

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Local.0.MathlibAnnex.IsometryEquiv.map_lineMap_of_map_midpoint`

The proof or construction of Affine-segment preservation on a quarter ball uses the project declaration “Continuous midpoint maps preserve affine segments” at the indicated step.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_mem_ball`

The proof or construction of Affine-segment preservation on a quarter ball uses the project declaration “Midpoint preservation on a quarter ball” at the indicated step.

Lean source signature (exact)

```
theorem map_lineMap_of_mem_ball
  {s : Set E} {t : Set F} (f : s  $\approx_i$  t) (c : s) {R :  $\mathbb{R}$ } (hR : 0 < R)
  (hsball : ball (c : E) R  $\subseteq$  s)
  (htball : ball ((f c : t) : F) R  $\subseteq$  t)
  (x y : s) (hx : (x : E)  $\in$  ball (c : E) (R / 4))
  (hy : (y : E)  $\in$  ball (c : E) (R / 4))
  (a :  $\mathbb{R}$ ) (ha : a  $\in$  Icc (0 :  $\mathbb{R}$ ) 1) :
  ((f (AffineMap.lineMap (x : E) (y : E) a,
    hsball (ball_subset_ball (by linarith : R / 4  $\leq$  R)
      ((convex_ball (c : E) (R / 4)).lineMap_mem hx hy ha))) : t) : F) =
  AffineMap.lineMap ((f x : t) : F) ((f y : t) : F) a
```

[Read exact MathlibAnnex v0.4.0 source](#)

Exact Card identity

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Approved exposition SHA-256: a0b1cd8f07e2ab200e5ad977cd661412712060d12dff543f6f52eee2aeec10a

AI-assisted material. Selected, edited, and reviewed by Ryotaro Tanaka. The recorded review is source-exposition correspondence, not an independent proof audit.